

Car-following Models: Data-driven Model

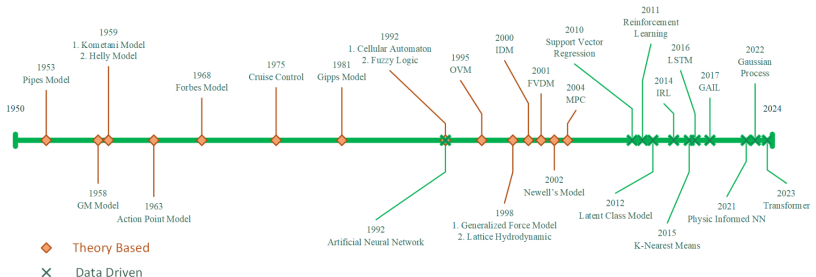
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<https://www.GoTrafficGo.com>

January 2, 2026

Introduction

Zhenbing He, Liang Zheng, Wei Guan (2015). A simple nonparametric car-following model driven by field data. *Transportation Research Part B: Methodological*, 80(2015), 185-201



Zhang et al, *Car-Following Models: A Multidisciplinary Review*, *IEEE Transactions on Intelligent Vehicles*, 2025

Existing *mathematical car-following model*, such as Gipps, IDM, Full velocity model

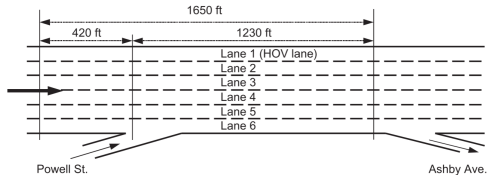
- Ordinary differential equation (ODE) with speed and acceleration variables
- Fundamental diagrams and driver's behavior parameters
- Calibration is needed before practical usage
- Should be able to reproduce major traffic characteristics

Contributions of this work:

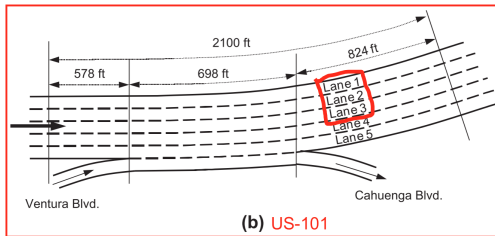
- A **data-driven** car-following model based on **k-nearest neighbor**
- Neither **mathematical equation** nor **calibration** is needed
- Neither **fundamental diagrams** nor **driver's behavior parameters** is assumed
- The model is **simple and parsimonious**, because there is only **one** parameter
- The model is able to **well reproduce** important traffic characteristics

Modeling

- Open-source: 15min+15min+15min trajectory dataset, 2005

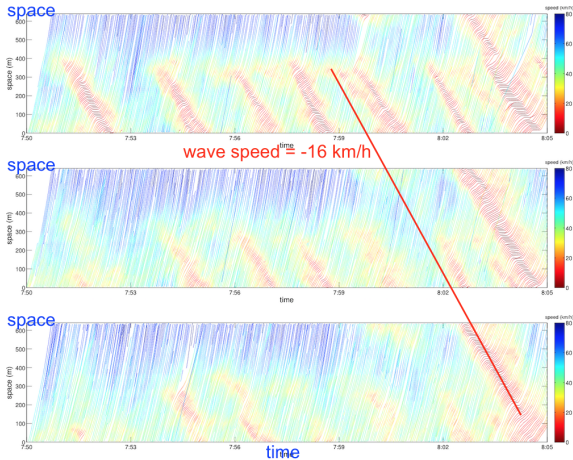


(a) I-80



(b) US-101

Data: NGSIM



Three Databases for Training

We build **three databases** with different sizes by using the datasets collected on **Lane 2 and 3**.

- Database 1: the small-size database containing 22,202 input-output samples, which is built by using the dataset collected on **Lane 2 during the first 15 minutes**
- Database 2: the medium-size database containing 78,683 samples, which is built by using the dataset collected on **Lane 2 during the all study 45 minutes**
- Database 3: the large-size database containing 152,637 samples, which is built by using the dataset collected on **Lane 2 and 3 during the all study 45 minutes**

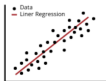
Such databases are employed to estimate the movement of the followers on **Lane 1**.

It aims to prevent the model from being influenced by Lane 1's data, ensuring objective evaluation and better generalization to unseen traffic scenarios.

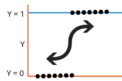
Nonparametric Approach: k -Nearest Neighbor

Top 8 Machine Learning Algorithms explained in less than 1 minute each

Linear Regression



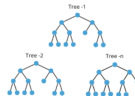
Logistic Regression



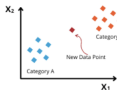
Decision Trees



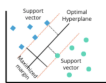
Random Forest



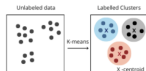
K-Nearest Neighbor



Support Vector Machine



K-Means Clustering



Naïve Bayes



Nonparametric Approach: k -Nearest Neighbor

Nonparametric approach/model

- Parametric approach: based on **mathematic formulas**
- Nonparametric approach: **no model**, driven by historical data

k -nearest neighbor: very simple but works well

- **History is repeating**: Most similar conditions (input) highly likely result in similar outcome (output).
- For example:
 - Wind, temperature, traffic $\rightarrow$ magnitude of haze.
 - Looking up similar days in history based on wind, temperature, traffic today
 - Taking **the average magnitudes of haze in these days** as the **estimator of the haze today**.

Nonparametric Approach: k -Nearest Neighbor

The approach selects **the most similar historical cases**, and takes the **average** of their outputs as the estimate of this time.

Specifically, the approach estimates $\mathbf{y}_0$ in focal $(\mathbf{x}_0, \mathbf{y}_0)$ as follows.

$$\hat{\mathbf{y}}_0 = \frac{\sum_{i=1}^k \mathbf{y}_i}{k} \quad (1)$$

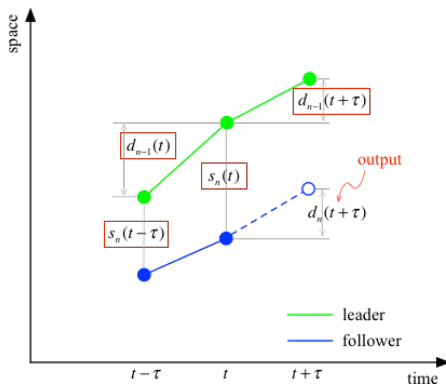
where $\mathbf{x}_i$ with respect to $\mathbf{y}_i$ is one of the k most similar samples to $\mathbf{x}_0$.

The Model: Key Assumption

Drivers tend to behave similarly in similar traffic situations.

The Model: Illustration

A key to understand the nonparametric model:



- Input (**traffic situation**): Leader's two-step movement + follower's two-step spacing $\rightarrow$ 1 step: speed, 2 steps: acceleration
- Output (**driver's behavior**): Follower's movement

The Model: Mathematical Expression

Input:

$$\mathbf{x}_n(t + \tau) = (d_{n-1}(t + \tau), d_{n-1}(t), s_n(t), s_n(t - \tau)). \quad (2)$$

where τ is the simulation time step; $(n - 1)$ is the leader of vehicle n ; $d_{n-1}(t)$ is the moving distance of the leader between time $(t - \tau)$ and t

Output: moving distance of vehicle n

$$\mathbf{y}_n(t + \tau) = d_n(t + \tau) \quad (3)$$

Measurement of Similarity

Distance between Two Data Samples

- “Ordinary” and simple **scaled Euclidean distance**, i.e., adjusting the input $x_{ji} \in \mathbf{x}_i$ by its mean $\bar{x}_j$ and standard deviation S_j before calculating Euclidean distance
- **The model reads**

$$\mathcal{D}(\mathbf{x}_i, \mathbf{x}_0) = \sqrt{\sum_{j=1}^J (z_{ji} - z_{j0})^2} \quad (4)$$

where

$$z_{ji} = \frac{x_{ji} - \bar{x}_j}{S_j}, \quad (5)$$

and J is the total number of all elements in an input vector.

Elimination of Autocorrelation

Because of **strong autocorrelation** of time-series trajectory data, it is easy for all k samples coming from a leader-follower pair, **in particular when the leader and follower move in a constant speed**. Such dominance could reduce the reliability of the model. To overcome this issue, we make all k samples selected from different leader-follower pairs.

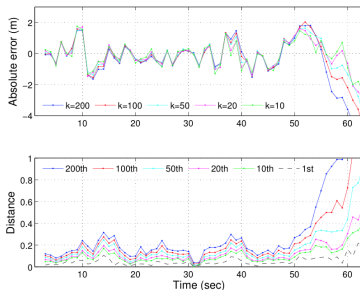
Determination of k

The only parameter in k NN

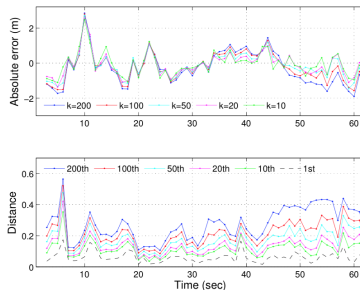
- k in k NN: the number of the historical cases that are considered to be similar to the estimated case.
- Usually, k is estimated by experience.
- Here, we determine k by comparing estimation errors under different k -values.
- Employ the scaled Euclidean distance from k th nearest sample to the estimated case, which is the longest distance in all k samples, Denote by $\mathcal{D}_k$.

Determination of k : First scenario

- We estimate the followers of two typical vehicles who traverse stop-and-go oscillations on Lane 1
- Database 3 is used, and the absolute errors with different k -values are compared



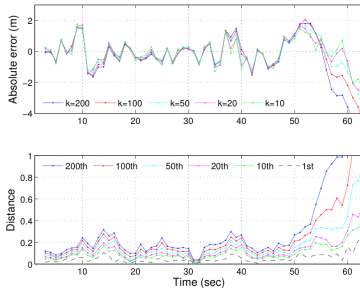
(a) Following Vehicle 422 on Lane 1



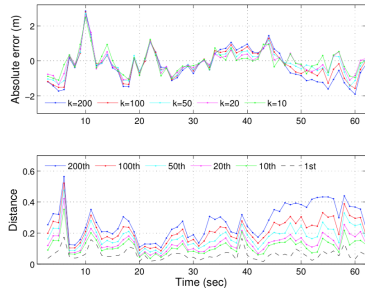
(b) Following Vehicle 1989 on Lane 1

Determination of k : First scenario

- **Upper plot (absolute errors):** smaller k basically makes smaller errors, but the errors may fluctuate more
 - the closer historical samples better reflect estimated case, but averaging a small number of samples results in instability
 - General tendency but not always true.



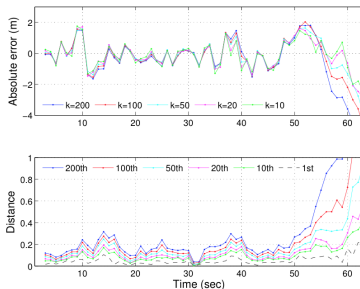
(a) Following Vehicle 422 on Lane 1



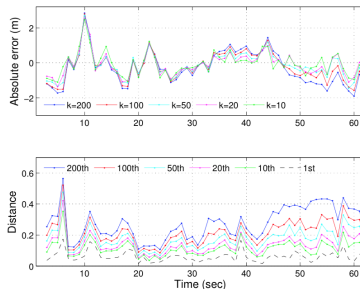
(b) Following Vehicle 1989 on Lane 1

Determination of k : First scenario

- Lower plot (distance $\mathcal{D}_k$): smaller k basically makes smaller distance (more similar)
 - For Database 3, when $k = 10$, the estimations are usually good with the distance smaller than 0.2, i.e. $\mathcal{D}_k < 0.2$



(a) Following Vehicle 422 on Lane 1

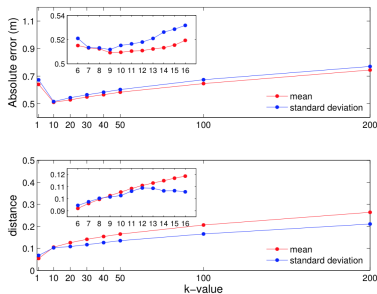


(b) Following Vehicle 1989 on Lane 1

Determination of k : Second Scenario

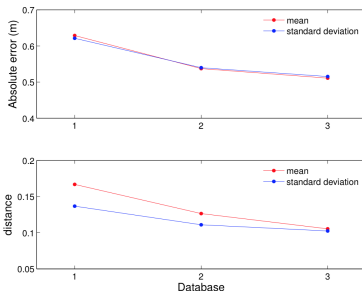
To make a more general conclusion,

- Using different k -values and Database 3, estimate **randomly-selected** 50 followers.
- Calculate the mean and standard deviation of absolute errors
- It can be seen:
 - Upper plot: for the dataset, the optimal k -value is about 10
 - Lower plot: usually, $\mathcal{D}_k < 0.2$



Determination of k : Second Scenario

- Fixing $k = 10$, and using different databases,
- It can be seen
 - The larger Database 3 results in better estimations (i.e., lower mean and standard deviation of the absolute errors)
 - Further shortens the distance $\mathcal{D}_k$.



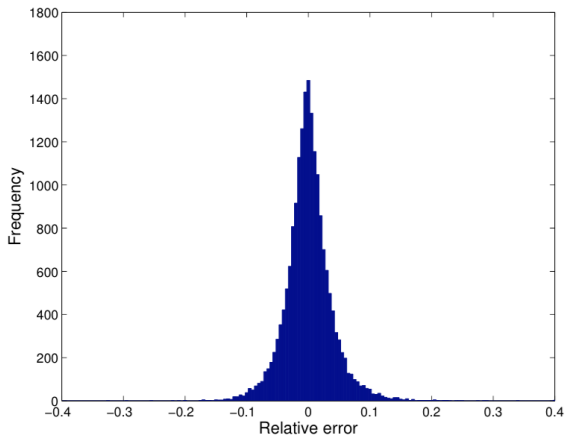
Determination of k : Conclusion

- Specify $k = 10$ for the database built on the US-101 dataset
- An estimation is considered to be satisfied if $\mathcal{D}_k < 0.2$
- This determination is a **premise to apply kNN** , because the optimal k -value is highly related to the underlying database

Analysis on Collisions

- Generally speaking, averaging similar collision-free historical cases would **not lead to a collision**.
- However, as a data-driven method, it may be **difficult to prove it mathematically**.
- To show collision-free **statistically**, we plot the relative errors between ground-true and estimated space headway.

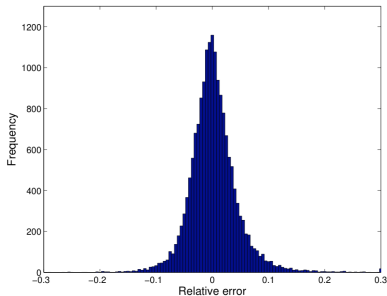
Analysis on Collisions



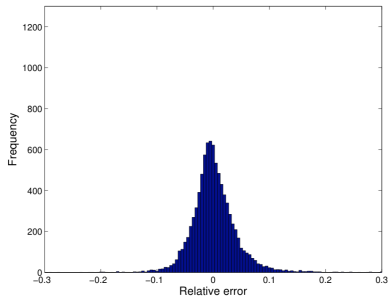
Transferability of the Model and the Database

- Basic assumption of k NN: drivers repeat their behavior in similar circumstances
- An database could be well transferred to any site with similar circumstances including driving habits, roadway geometry, etc
- To show this, we estimate vehicles on I-80 using US-101 database.

Transferability of the Model and the Database



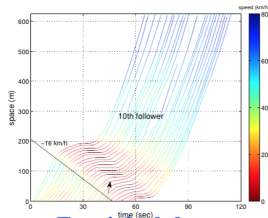
(a) Lane 2, I-80



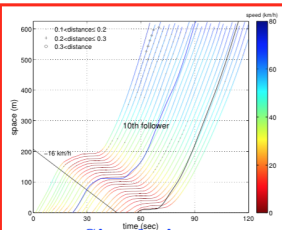
(b) Lane 3, I-80

Validation

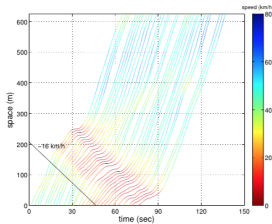
Scenario 1: A Platoon with Real Boundary Conditions



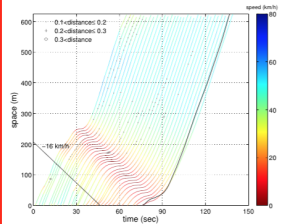
(a) **Empirical data**
Real platoon following Vehicle 422



(b) **Simulation**
Simulated platoon following Vehicle 422

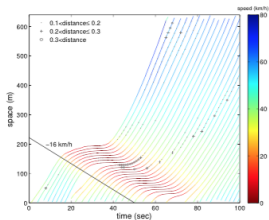


(c) Real platoon following Vehicle 1989

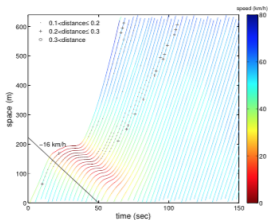


(d) Simulated platoon following Vehicle 1989

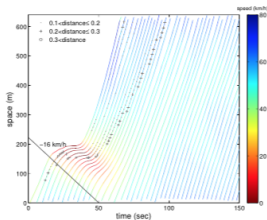
Scenario 1: A Platoon with Real Boundary Conditions



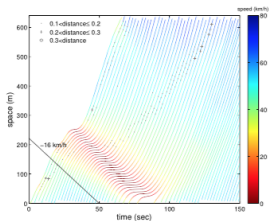
(a) Vehicle 422, entry gap 30 m



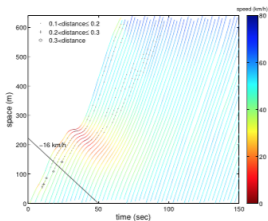
(b) Vehicle 422, entry gap 40 m



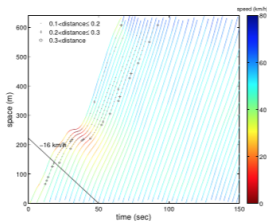
(c) Vehicle 422, entry gap 50 m



(d) Vehicle 1989, entry gap 30 m

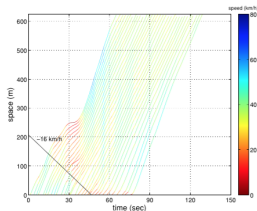


(e) Vehicle 1989, entry gap 40 m

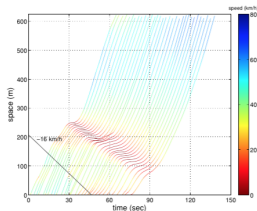


(f) Vehicle 1989, entry gap 50 m

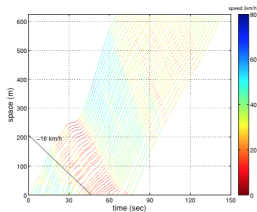
Ablation Experiment: Necessity of Each Input



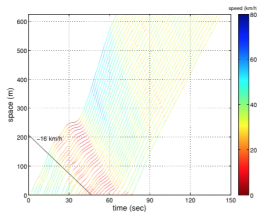
(a) Inputs without $d_{n-1}(t - \tau)$



(b) Inputs without $d_{n-1}(t)$



(c) Inputs without $s_n(t - \tau)$

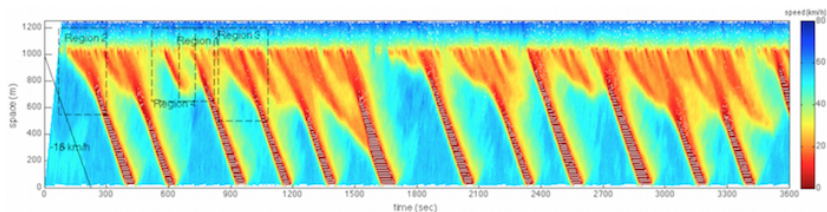


(d) Inputs without $s_n(t)$

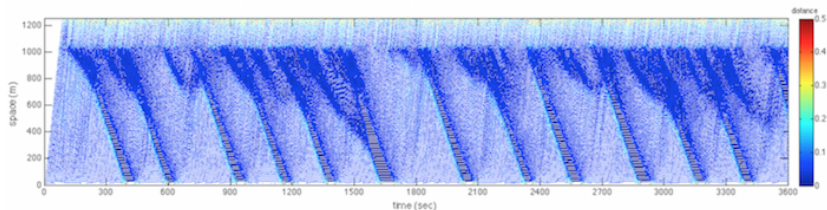
Scenario 2: Rubbernecking

- A 1.25 km one-lane roadway is simulated for 1 hour;
- New vehicle enters with an initial speed of 54 km h^{-1} , when its leader has left the entrance 30 m away;
- The rubbernecking zone is located at section $[1, 1.05] \text{ km}$;
- A probability r to rubberneck and then slow down by a percentage of $(1 - p)$;
- If rubbernecking occurs, it will occur at most once;
- The database contains all data collected from Lane 1, 2, and 3 during all 45 minutes;
- No assumption or calibration

Simulation Results: Time-Space Plots

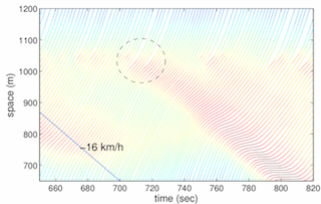


(a) Time-space diagram coloured by speed

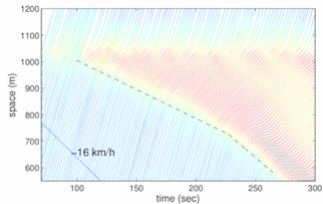


(b) Time-space diagram coloured by distance

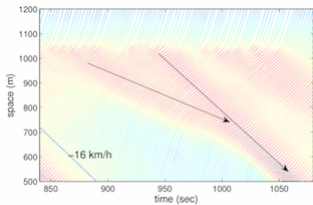
Simulation Results: Time-Space Plots



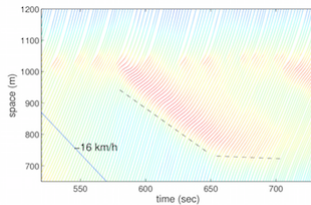
(c) Region 1



(d) Region 2



(e) Region 3



(f) Region 4

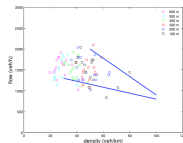
Simulation Results: Fundamental Diagrams

Suppose that **virtual detectors** are installed in the roadside, and the traffic flow q , density ρ , and speed v within a time period T are measured as standard models.

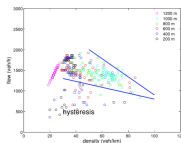
$$q = \frac{N}{T}, \quad \rho = \frac{\sum_{n=1}^N \frac{1}{v_n}}{T}, \quad \text{and} \quad v = \frac{q}{\rho} = \frac{N}{\sum_{n=1}^N \frac{1}{v_n}} \quad (6)$$

where N is the count of the vehicles passing the detection location within the time period T , and v_n is the passing speed of a detected vehicle.

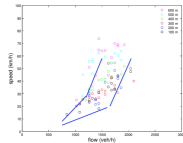
Simulation Results: Fundamental Diagrams



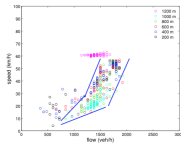
(a) flow-density diagram for empirical traffic



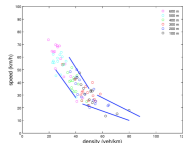
(b) flow-density diagram for simulated traffic



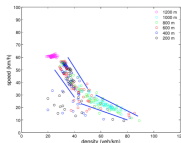
(c) speed-flow diagram for empirical traffic



(d) speed-flow diagram for simulated traffic

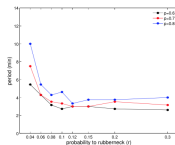


(e) speed-density diagram for empirical traffic

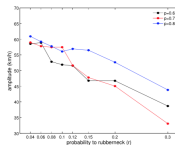


(f) speed-density diagram for simulated traffic

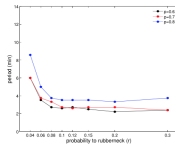
Simulation Results: Periods and Amplitudes



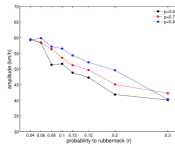
(a) Period ($h=4$)



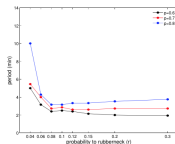
(b) Amplitude ($h=4$)



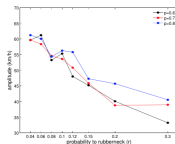
(c) Period ($h=5$)



(d) Amplitude ($h=5$)



(e) Period ($h=6$)



(f) Amplitude ($h=6$)

Scenario 3: Driver Errors

We model the driver errors in a form of a **white Gaussian noise** with diffusion coefficient σ^2 .

$$\tilde{d}_n(t + \tau) = d_n(t + \tau) + W(\varepsilon) \quad (7)$$

where

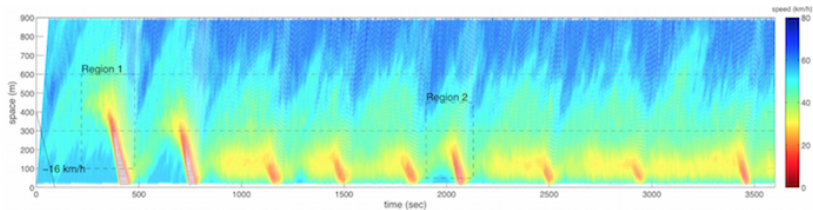
$$W(\varepsilon) = \begin{cases} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{\varepsilon^2}{2\sigma^2}} & , \quad d_{\text{jam}} < d_n(t + \tau) < d_{\text{free}} \\ 0 & , \quad \text{otherwise} \end{cases} \quad (8)$$

where d_{jam} and d_{free} are the moving distance/speed around jam density and free-flow conditions, respectively.

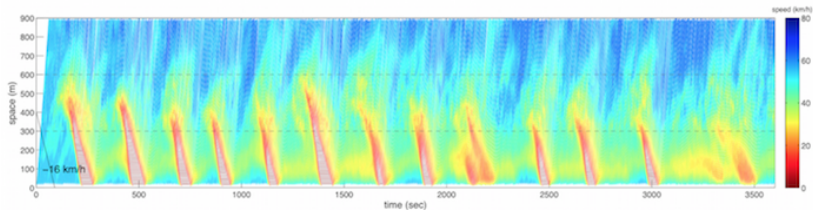
Scenario 3: Driver Errors

- A 900 m one-lane roadway is simulated for 1 h;
- The white noise is only added when a vehicle is moving in the section between 300 m and 600 m. This is analogous to an uphill section;
- It is set that $d_{\text{free}}=54 \text{ km h}^{-1}$ and $d_{\text{jam}}=15 \text{ km h}^{-1}$;
- The entry speed and gap are 54 km h^{-1} and 20 m, respectively.

Simulation Results: Time-Space Plots

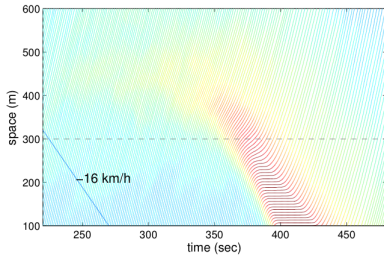


(a) Time-space diagram of trajectories ($\sigma = 0.2$)

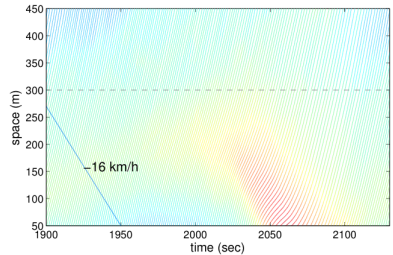


(b) Time-space diagram of trajectories ($\sigma = 0.5$)

Simulation Results: Time-Space Plots



(c) Region 1



(d) Region 2

Conclusion

- Neither mathematical equation nor calibration is needed to be concerned in the model;
- Neither the fundamental diagrams nor driver's behaviour parameters is assumed;
- The model is simple and parsimonious particularly in the conceptual point of view, and the only parameter is k ;
- All inputs and outputs are based on vehicle positions, which are straightforward to reproduce traffic dynamics in computer simulations;
- The model is able to well reproduce traffic characteristics contained by the underlying database, such as all stages of stop-and-go oscillations, fundamental diagrams, periods and amplitudes of oscillations.

Thank you!